

TIME ALLOWED 85 MINS

INSTRUCTIONS:

All necessary working should be shown.
This is an OPEN BOOK test
Each section should be started on a new page.

Section A (Start a new page)

- (a) A particle moves in a straight line in such a way that its distance in metres from the origin after t secs is given by $x = 2t^3 + 3t^2 - 36t + 10$.
- Find the velocity and acceleration at any time t
 - Find its initial position.
 - In which direction is the particle initially moving? Give a reason.
 - When and where does it come to rest?

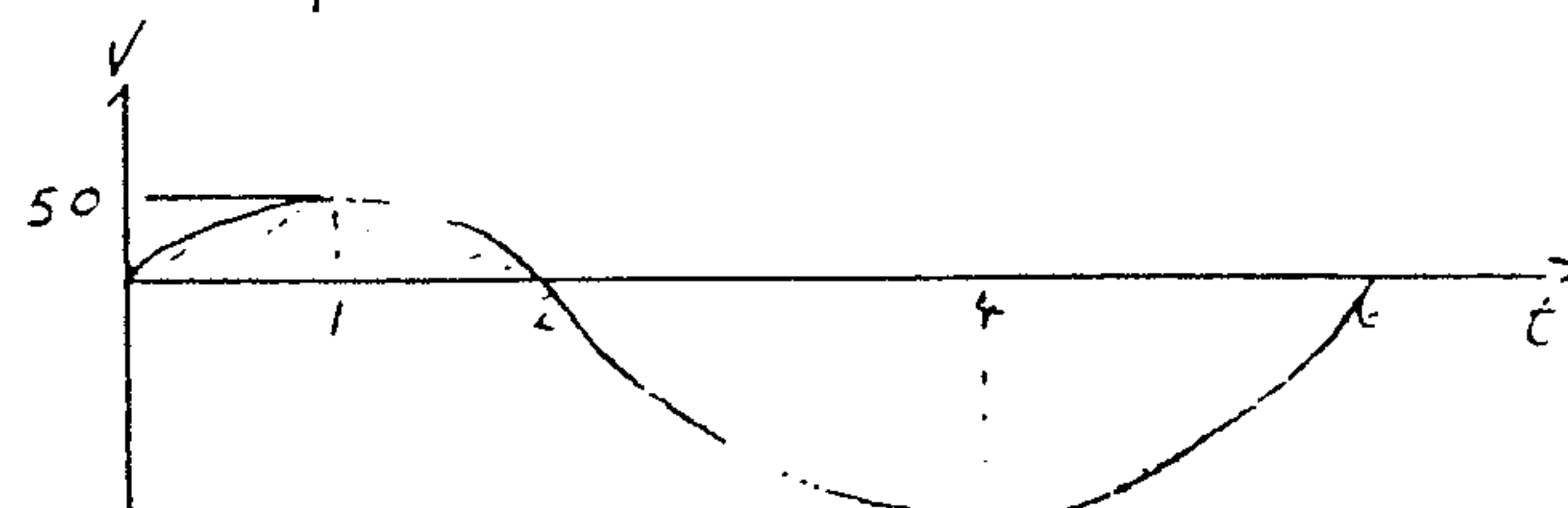
- (b) A bag contains green, black and red jellybeans. Therefore if I choose one jellybean at random from the bag, the probability that it is black is $\frac{1}{3}$. Is this statement true or false? Explain why, in no more than one sentence.

- (c) A box contains 5 black, 7 red and 4 white marbles. One marble is drawn. Find the probability that:
- it is red or white
 - it is not white

Section B (Start a new page)

- (a) A particle moves in a straight line in such a way that its velocity in ms^{-1} at time t secs is given by $v(t) = at^2 - bt + c$ where a, b and c are constants. Knowing $v(1) = 1$, $v(2) = 9$ and $v(0) = 3$:
- find a, b and c
 - by considering the discriminant show that the particle never comes to rest.

- (b) The graph represents the velocity v m/s of a particle after t seconds travelling in a straight line. The particle starts from rest.



- What is the velocity of the particle after 1 second?
- When does the particle change direction?
- When is the acceleration of the particle zero?
- What happens to the particle after 6 seconds?
- Explain what is represented by the shaded region in the diagram

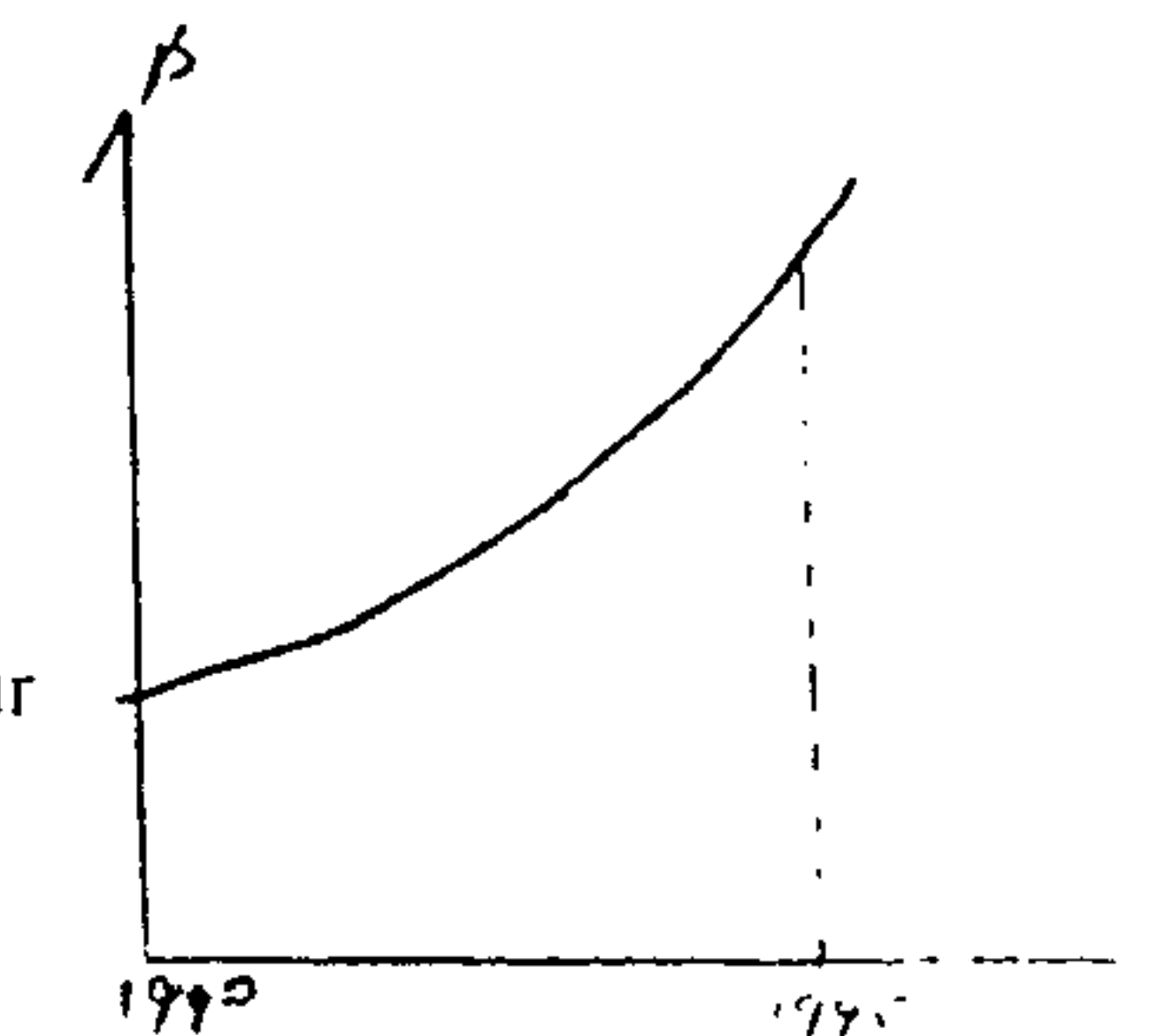
Section C (Start a new page)

- (a) The population of kangaroos on an island is decreasing according to the formula $N = N_0 e^{-kt}$ where N is the kangaroo population and k is a constant, t is time in years.
- If the population at the end of 1980 was 7000 and at the end of 1990 was 4000, show that the value of k is 0.0560 correct to 4 decimal places.
 - Find in how many years (from 1980) will it take for the population to be 40% of the original population
 - What will the population be after 5 years?
 - What is the rate of change at $t = 5$?

- (b) A box contains 5 black, 3 white and 2 pink plastic discs. By drawing a tree diagram or otherwise find the probability that, if two are drawn in succession without replacement:
- both are the same colour
 - one is black and the other white

Section D (Start a new page)

- (a) The graph shows the population of a particular beetle between 1990 and 1995.



- Briefly describe how the beetle population has changed over the period (include the rate of change of the population in your discussion)

- Comment on the sign of $\frac{d^2B}{dt^2}$

- (b) A second type of insect (I) has the same population in 1990 as the beetle in (a).

Between 1990 and 1995 (inclusive) $\frac{dI}{dt} < 0$

- What does this say about the number of this second type of insect during the period 1990-95?

- It is also observed that in the period 1990-95, $\frac{d^2I}{dt^2} > 0$

Sketch I against t for the time period 1990-95.

- (c) The probability that a particular type of shrub will flower in the first year after planting is $\frac{1}{3}$

- Find the probability that:
 - a shrub will not flower
 - if three shrubs are planted all three will not flower
 - if three shrubs are planted at least one will flower
- Show that if 9 shrubs are planted then the probability of having at least one flowering plant is greater than 90%.

Section E (Start a new page)

- (a) The acceleration $a \text{ ms}^{-2}$ of a particle is given by $a = 4\pi^2 \cos 2\pi t$. The particle starts from rest at the origin. Find:

- the velocity and displacement at any time t
 - Draw a graph of x against t for $0 \leq t \leq 3$
 - A second particles position is give by $x = 12 - 4t + t^2$
- Show that the two particles never meet.

- (b) (i) Two dice are thrown together. Find the probability that:

- the sum of the dice is 4
- the sum of the dice is 6
- the sum of the dice is neither a 4 nor a 6.

- (ii) If the pair of dice are thrown twice find the probability that neither a sum of 4 nor a sum of six is thrown on the first throw and a sum of four is thrown on the second throw.

- (iii) Find the probability that a sum of 4 is thrown before a sum of 6 (considering an unlimited number of throws). Give answer in simplified form.

(i) $x = 2t^3 + 3t^2 - 76t + 10$
 $V = 6t^2 + 6t - 76$
 $a = 12t + 6$
 (ii) $x = 10$ when $t = 0$
 (iii) $V = -36$ when $t = 0$
 ... moving left
 (iv) when $v = 0$
 $0 = 6(t^2 + t - 12)$
 $= 6(t+4)(t-3)$
 $t = 2$ only
 $x = 2.8 + 3.4 - 36.2 + 10$
 $= -34$

FALSE: there is no guarantee that the numbers of each colour are equal

(i) $\frac{7}{16} + \frac{4}{16} = \frac{11}{16}$
 (ii) $1 - \frac{4}{16} = \frac{12}{16}$

Section B (13 marks)

$v(t) = at^2 - bt + c$
 $1 = a - b + c$
 $9 = 4a - 2b + c$
 $c = 3$
 $-2 = 6a - b$
 $b = 4a - 2b$
 $4 = 2a - 2b$
 $2a = 10$
 $a = 5$
 $b = 7$

$s(t) = 5t^2 - 7t + 3$
 $s = -11$
 1/0: Normal solⁿ for t
 : Never comes to rest.

(b) (i) $V = 50$
 (ii) $t = 2$ (NOT $t = 1$)
 (iii) $t = 1, 4$
 (iv) particle stops
 (v) Particle travels (pos. direction)

Section C (14 marks)

(a) $N = N_0 e^{-kt}$
 (i) $4000 = 7000 e^{-10k}$
 $\ln \frac{4}{7} = -10k$
 $-k = 0.0560$ (4 d)
 (ii) $\frac{4}{10} = e^{-0.0560t}$
 $\ln \frac{4}{10} = -0.0560t$
 $t = 16.4$ yrs

(iii) $N = 7000 e^{-0.0560 \times 15}$
 $N = 5291.5$

(iv) $\frac{dN}{dt} = -kN e^{-kt}$
 $= -0.0560 \times 7000 e^{-0.0560 \times 15}$
 $= -296.12$
 decreasing at 296 kg/yr

(v) $\frac{5}{15} \cdot \frac{4}{7} + \frac{3}{15} \cdot \frac{4}{7} + \frac{2}{15} \cdot \frac{4}{7}$
 $= \frac{14}{45}$

(vi) $2 \cdot \frac{5}{10} \cdot \frac{2}{7}$
 $= \frac{1}{3}$

Section D (13 marks)

(a) (i) Population has increased (at an ever increasing rate)
 (ii) $\frac{d^2P}{dt^2} > 0$

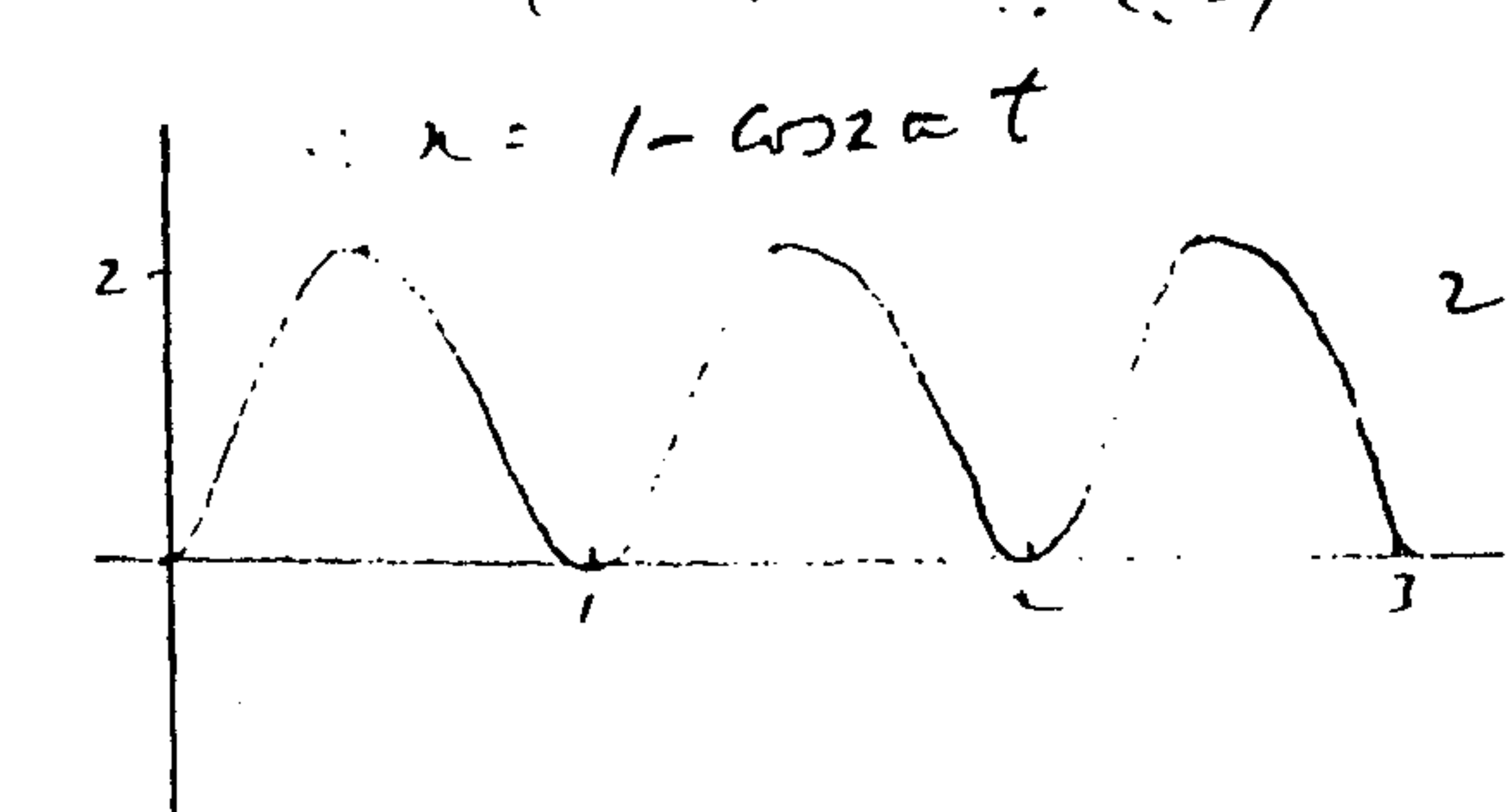
(b) (i) Number is decreasing



(c) (i) (a) $\frac{3}{4}$
 (b) $(\frac{3}{4})^3$
 (ii) $1 - (\frac{3}{4})^3$

Section E (17 marks)

(a) $a = 4\pi^2 \cos 2\pi t$
 $v = 2\pi a \sin 2\pi t + c_1$
 $t = 0, v = 0 \therefore c_1 = 0$
 $x = -\cos 2\pi t + c_2$
 $t = 0, x = 0 \therefore c_2 = 1$



Consider $x = 12 - 4t + t^2$
 Min value of 8 when $t = 2$
 Does not meet $y = 1 - \cos 2\pi t$ which has max value of 2

(i) (a) $\frac{1}{12}$
 (b) $\frac{5}{36}$
 (ii) $\frac{28}{36} \times \frac{1}{12}$
 (iii) $\frac{3}{8} \left[\frac{1}{12} + \frac{28}{36} \cdot \frac{1}{12} + \left(\frac{28}{36}\right)^2 \cdot \frac{1}{12} + \dots \right]$